

# Leakage-Aware Multi-Horizon Benchmarking of Compact Neural PV Forecasts Across Technologies and Sites

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Photovoltaic forecasting rankings depend on the information available to a model, the targets being scored, and the reference forecast. We examine these dependencies across three geographic sites using a two-level benchmark: three co-located arrays at Alice Springs with 17 input channels, and independently fitted replications at Yulara and NIST Ground with seven common channels. Four compact neural implementations use six hours of history to predict a twelve-hour trajectory, evaluated at 1, 4, 8, and 12 h on full-timeline and daylight scopes. The study comprises 60 training runs: 36 for Alice Springs in 2018 and 24 for the external sites in 2017. Train-only preprocessing, split-local windows, Validation-only checkpoint selection, and identical target masks define each comparison. At Alice Springs, Daily Persistence has lower error than the post hoc best-of-four neural envelope in 22 of 24 comparisons. At the external sites, the prespecified Inverted-variate model improves on Last-value Persistence in all 16 comparisons and on matched Daily Persistence in 15. Its first-place ranking at Yulara changes at NIST, where recurrent and depthwise implementations lead different evaluation scopes. These results distinguish improvement beyond local continuity from improvement beyond daily recurrence. They support a benchmark interpretation in which site, horizon, scope, and historical information jointly condition the observed advantage.

Keywords: photovoltaic forecasting, multi-horizon benchmark, sample matching, persistence, external-site replication

## I. INTRODUCTION

Photovoltaic (PV) forecasts support storage scheduling, distribution operation, and reserve allocation. A useful benchmark should establish what a forecast adds to the observations already available at its issue time. This question depends on horizon, weather information, target definition, and the reference method. Reviews of PV forecasting therefore emphasize the context needed to interpret reported errors.<sup>1–3</sup> Recurrent, convolutional, and attention-based models provide several ways to summarize historical observations, but an architecture name alone does not specify a forecasting experiment.<sup>4,5</sup>

Three evaluation choices are central. Transformations fitted beyond the training period can expose a model to later distributions. Missing labels and lagged observations can give competing methods different targets. A reference that repeats the present value asks a different question from one that retrieves yesterday’s value. Temporal evaluation guidance and leakage studies motivate explicit controls for these choices.<sup>6–9</sup> In PV forecasting, their interaction matters because nighttime production, daytime variability, and daily recurrence contribute differently to an aggregate error.

Public PV archives also permit two complementary forms of replication. Co-located arrays share much of

their weather context while retaining different electrical systems and power targets. Geographically separate facilities test whether an observed pattern appears under another operating distribution. Alice Springs provides the first setting through its documented technology arrays.<sup>10–12</sup> Yulara and NIST Ground provide the second through separate power and meteorological records.<sup>13,14</sup> These settings need not have identical sensor inventories: the information available to each experiment can be made explicit instead of treating all inputs as interchangeable.

We combine these two levels in a history-only benchmark. The Alice Springs experiment compares three co-located arrays with 17 channels. The external-site replications use the common seven-channel information set available at Yulara and NIST and fit each model separately at each site. All settings use six-hour histories, direct twelve-hour trajectories, four evaluation horizons, two scopes, and two persistence references. The Inverted-variate implementation was fixed as the external primary model from the frozen Alice results before external Test predictions were generated. The four-model external comparison remains secondary.

The study asks whether improvement beyond Last-value Persistence recurs at the external sites, whether Daily Persistence changes that conclusion, and whether model rankings remain consistent across arrays, horizons, and scopes. Its contribution is a two-level empirical benchmark with explicit information sets and matched evaluation targets. The strongest contrast is between daily recurrence at Alice Springs and the external primary model’s gains over both references. The accompanying ranking changes identify the conditions under

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which a particular implementation leads, rather than assigning one architecture a universal position.

## II. RELATED WORK

### A. PV forecasting models and information availability

PV forecasting spans physical, statistical, machine-learning, and hybrid methods. Historical power and ground meteorology are commonly used for intra-hour and short intraday prediction; future numerical weather prediction (NWP), satellite products, and sky imagery become increasingly important as the horizon grows.<sup>1,2</sup> This distinction matters because a history-only model and an NWP-assisted model estimate different conditional quantities. Cross-Unet, for example, explicitly combines historical observations with forward weather fields, demonstrating the value of a richer forecast-time information set rather than providing a directly comparable error for a history-only task.<sup>15</sup>

Recent work has also broadened the scale and objectives of benchmarking. SolNet supplies open-source models across geographically distributed systems and emphasizes transfer across regions.<sup>16</sup> Markovics *et al.* compare classical and deep models using weather forecasts, extensive tuning, and multiple sites.<sup>17</sup> Dhingra *et al.* examine neural forecasts under data constraints and include uncertainty quantification, while related work by the same group studies quantile forecasts under data loss.<sup>18,19</sup> A 2025 review identifies standardization, generalization, and uncertainty as recurring open problems.<sup>3</sup> These studies address geographic transfer, future weather information, or uncertainty. Here, two input regimes distinguish a co-located technology comparison from independently fitted external-site replications, with target support matched within each setting.

Central Australian PV archives have already supported several forecasting studies. Mahmud *et al.* used machine learning for power forecasting at Alice Springs, Huang *et al.* evaluated a neural-ODE-based architecture on three Alice Springs arrays, and Mansour *et al.* benchmarked six machine-learning models against Persistence on high-resolution Yulara data under seasonal and stress conditions.<sup>12,20,21</sup> These studies establish the relevance of public desert-PV data. Data availability, split definitions, persistence lags, model inputs, and eligible samples differ across studies; consequently, reported numerical errors are not directly comparable.

### B. Compact time-series mechanisms

Long short-term memory networks and temporal convolutional networks are established sequence models.<sup>22,23</sup> Transformer variants introduced efficient attention, decomposition, frequency-domain structure, patch tokenization, and inverted variate tokens.<sup>24–29</sup> ModernTCN

revisited convolutional time-series modeling with large kernels and modern block design.<sup>30</sup> These papers define architecture families, not the exact compact implementations evaluated here.

Our four forecasters retain one recognizable mechanism from those families under a common input-output budget: recurrent trajectory decoding, variate tokenization, joint temporal patching, or depthwise temporal convolution. The resulting networks are compact project implementations rather than full official reproductions. Their rankings concern these implementations under the stated information and training budgets.

### C. Persistence, leakage, and reproducible evaluation

Forecasting benchmarks require naive references whose information sets are explicit.<sup>31,32</sup> Persistence is particularly competitive for solar generation because power is smooth over minutes and approximately periodic over days. Skill relative to persistence communicates whether a forecast adds value beyond that structure, but only if the reference and model share targets and masks. Daily lag availability can otherwise create a hidden sample-selection effect.

Temporal leakage can enter through preprocessing, windows, or model selection. Random cross-validation is inappropriate for many dependent series, and globally fitted transformations can make a nominally chronological evaluation transductive.<sup>7,8</sup> Reproducibility further requires definitions of split boundaries, target timestamps, inverse scaling, checkpoint selection, and metric aggregation. Three random seeds cannot establish universal statistical superiority, but reporting them prevents a favorable initialization from being silently selected.

## III. STUDY DESIGN AND DATASETS

### A. Two levels of evidence

Figure 1 separates the co-located comparison from the external-site replications. Each array or external facility has its own fitted preprocessing and model weights. No data are pooled across sites. The original 36 runs and the external 24 runs use the same four compact mechanisms and seeds 42, 43, and 44, with different input widths determined by the available fields. Table I records the corresponding dates, capacities, and time coordinates.

### B. Alice Springs: co-located technology comparison

The Alice Springs targets are Site 17 Sanyo HIT-210NKHE5 hybrid-silicon modules, Site 25 Hanwha HSL 60S polycrystalline-silicon modules, and Site 38 Q CELLS Q.PEAK-G4.1 monocrystalline-silicon

## Two evidence levels; independent fits for every array or facility

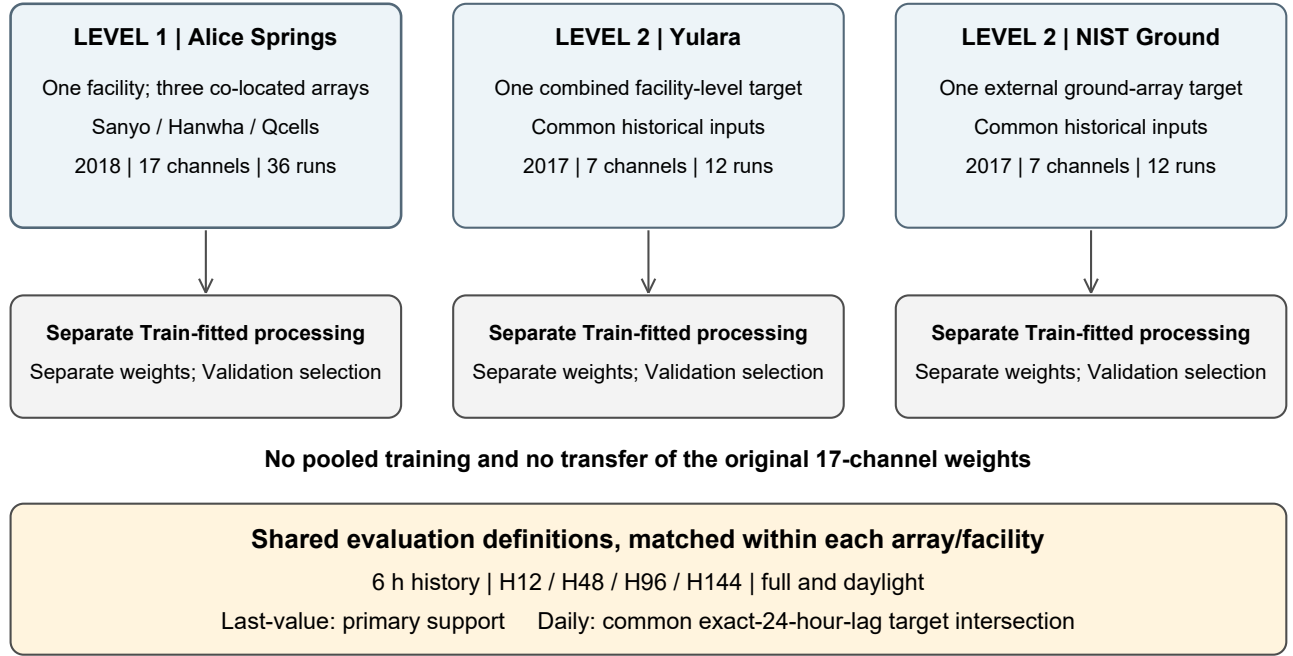


FIG. 1. Two-level study design. Alice Springs contributes three co-located technology arrays with 17-channel inputs. Yulara and NIST each contribute one external facility with seven-channel inputs. Every target system is fitted separately; the common element is the window and matched-evaluation protocol, not a pooled model.

TABLE I. Three geographic sites and two input regimes. Capacities are descriptive DC ratings; error normalization uses each target’s Train range. All forecast grids are five-minute grids. Dates in the last two rows refer to 2017, and those in the first row to 2018.

| Site / systems                          | DC rating / channels                                  | Train; Validation; Test              | Source resolution / time basis                | Interpretation limit                                    |
|-----------------------------------------|-------------------------------------------------------|--------------------------------------|-----------------------------------------------|---------------------------------------------------------|
| Alice Springs / three co-located arrays | Sanyo 6.30, Hanwha 5.83, Qcells 5.90 kW / 17 channels | Apr 1–Jul 15; Jul 16–Aug 7; Aug 8–31 | 5 min / frozen ACST coordinate                | Shared weather; technology is not an isolated treatment |
| Yulara / one combined facility          | 106.6 kW / 7                                          | Jan–Aug; Sep; Oct–Dec                | 5 min / provider local; +5 min availability   | GHI metadata and UTC offset partly unconfirmed          |
| NIST Ground / one array                 | 271 kW / 7                                            | Jan–Aug; Sep; Oct–Dec                | 1 min to 5 min / fixed EST, UTC–05:00, no DST | Complete 2017 operation log unavailable                 |

modules.<sup>10</sup> Their historical Active Power and Performance Ratio are array-specific. Ambient temperature, relative humidity, global and diffuse horizontal radiation, and global and diffuse tilted radiation supply six shared weather fields. Together these form eight numerical channels, eight original-missingness masks, and one Isolation Forest flag: 17 channels in total.

The existing April–August 2018 protocol is retained. The target is AC active power in kW. The frozen Alice validity rule excludes negative power labels; the external rule below retains finite negative meter power. This

difference is part of the two experimental regimes, alongside the years, sensors, and targets. Co-location provides common weather context but does not identify a causal effect of cell technology: orientation, balance-of-system behavior, and sensing remain relevant differences.

### C. Yulara: combined-system replication

The Yulara record represents the 106.6 kW Sails in the Desert rooftop combined system, installed

in 2016. Its component subarrays are treated as one facility-level target.<sup>13</sup> The downloaded file spans April 2016–February 2026; the experiment uses 2017. The three numerical fields are `Active_Power`, `Weather_Temperature_Celsius`, and `Global_Horizontal_Radiation`.

The 2017 file segment contains all 105,120 regular five-minute timestamps plus two off-grid records. The latter are excluded without rounding or overwriting their neighboring regular records. The provider local coordinate is preserved; an unconfirmed UTC offset is not assigned. Each value becomes available at its raw timestamp plus five minutes. This conservative rule prevents an ambiguous interval label from making a measurement available before its full interval has ended. The local coordinate is fixed and has no introduced DST transition.

Temperature is used as the Celsius variable named by the provider. GHI is used in its provider-native numerical form without additional conversion. Detailed sensor, aggregation, and invalid-code metadata are incomplete; native GHI magnitudes are therefore not used for absolute cross-site irradiance inference. Missing and non-finite entries remain missing, while finite negative observations are retained. The Supplement gives the excluded records, field statistics, and metadata limits.

#### D. NIST Ground: fixed-standard-time replication

The NIST Ground array in Gaithersburg supplies AC meter power and ground-level meteorological observations. Its documented DC rating is 271 kW.<sup>14</sup> The target and historical power are `PwrMtrP_kW_Avg`, the real power measured by the AC meter; temperature is `AmbTemp_C_Avg`; GHI is `Pyra1_Wm2_Avg`. The downloaded CSV already supplies the irradiance in W/m<sup>2</sup>, so no mV conversion is applied. The inverter-power column, including its −999 entries, is outside both the target and the common inputs. The field dictionary and official data catalogue establish the source lineage.<sup>33,34</sup>

The 365 daily files contain 525,595 unique ordered minutes. Five missing minutes straddle October 20–21. Time is retained as a timezone-aware fixed UTC−05:00 coordinate throughout parsing, aggregation, split assignment, and lag joins, consistent with the provider’s local standard time.<sup>14</sup> No daylight-saving adjustment is introduced. For each five-minute end time  $T$ , a variable is averaged over  $[T - 5 \text{ min}, T)$  and made available at  $T$ . The bin must contain five distinct minutes with finite values for that variable; otherwise its aggregate remains missing. Midnight anchors, left-closed bins, and right-end labels are explicit.

Train-period energy-increment and power–GHI timing diagnostics support the time alignment; they do not select rules from prediction errors. The complete 2017 operation log was not obtained. Accordingly, finite meter power, temperature, and GHI are retained without heuristic removal of possible outages, snow, maintenance,

or high-irradiance/low-power periods. Missingness is explicit, and the absence of a log is not interpreted as an absence of events.

## IV. COMPACT MODELS AND MATCHED EVALUATION

### A. Inputs, preprocessing, and forecast windows

The external numerical-channel order is historical AC power, ambient temperature, and GHI. The next three channels are their original missingness masks, followed by the Isolation Forest flag. Missingness is recorded before imputation. A five-neighbor KNN imputer is fitted on Train numerical inputs. Isolation Forest is then fitted on the imputed Train inputs with 100 trees, contamination 0.01, and random state 42.<sup>35,36</sup> The masks and anomaly flag are appended, and a MinMaxScaler is fitted to the resulting seven Train channels. A separate target scaler is fitted to finite, unimputed Train power. Validation and Test use these fitted transformations unchanged. Labels are never imputed. Alice uses its frozen eight-variable counterpart of this preprocessing sequence.

All regular timeline positions are retained. Missing records do not become adjacent to distant observations by deleting rows. At origin  $t$ , the input contains 72 five-minute positions ending at  $t$ ; every input value’s availability time is at most  $t$ . Targets start at  $t + 5 \text{ min}$  and extend to  $t + 720 \text{ min}$ . Input and target windows remain within one chronological split. No cyclical clock variables, future weather, humidity, or Performance Ratio are added to the external common inputs.

External Train covers January–August 2017, Validation September, and Test October–December. Boundaries use the fixed time coordinate of each source. These three Test months define the external held-out performance split. They were inspected for data-quality auditing, but were excluded from fitting preprocessing, model weights, and checkpoint selection. They are not a full-year seasonal evaluation.

### B. Four compact implementations and checkpoint selection

The Discrete recurrent implementation combines multi-dilation convolution, a compact Transformer branch, gated fusion, a bidirectional GRU summary, and a recurrent trajectory decoder. The Inverted-variate implementation treats each channel’s 72-point history as a token, applies a compact attention encoder, and maps its joint representation to H144. Joint-patch encodes overlapping multivariate temporal patches. Depthwise TCN uses an input projection, four depthwise/pointwise temporal blocks, and a direct trajectory head. These mechanisms draw on recurrent, variate-token, patch, and convolutional forecasting designs.<sup>22,28–30</sup> The complete

layer specifications and parameter counts for both input widths appear in the Supplement.

The 17- and seven-channel networks use the same compact structures with the required input and output projections. Seven-channel weights are fitted separately, not converted from the original checkpoints. The comparison holds the information set and optimization budget constant within each experiment; parameter counts differ with the implemented mechanism. Thus the results describe these project implementations rather than an official architecture leaderboard or a pure architectural causal effect.

Each of the 60 runs uses one of three fixed seeds and AdamW with learning rate 0.001, weight decay  $10^{-5}$ , batch size 256, gradient clipping 1.0, a maximum of 25 training passes, and early-stopping patience 5.<sup>37</sup> Minimum improvement is  $10^{-8}$ . Training and checkpoint selection use complete-H144 target windows. External windows additionally require a finite true origin power. Validation loss is total squared error divided by total valid target count, rather than an unweighted mean of batch losses. The best Validation checkpoint is fixed before Test prediction; no Test score selects a seed, model, or budget.

Test evaluation uses horizon-specific eligibility at H12, H48, H96, and H144. A valid H12 origin remains eligible even if a later target in steps 13–144 is missing. All models output H144, and evaluation reads the requested prefix. The Alice complete-H144-origin sensitivity is retained in the Supplement. This separates the common training objective from each requested forecast horizon’s evaluation support.

### C. Persistence references and scopes

Last-value Persistence repeats the true valid power at the forecast origin; it never uses an imputed origin value. Daily Persistence retrieves the true power at each target’s exact timestamp minus 24 h. This is a timestamp join in the frozen coordinate, not a row-number shift. Daily’s finite-lag mask is intersected with the target mask and applied elementwise to Daily, Last-value, and every neural seed. Primary neural/Last-value comparisons and supplementary Daily-matched comparisons therefore each have their own explicitly shared support.

The full-timeline scope includes every valid target. Daylight is defined by true future power exceeding 1% of the corresponding Train maximum. The threshold is fitted on Train, but scope membership uses labels after prediction; it is a descriptive evaluation scope rather than a deployable daylight detector. Both masks and thresholds are model-independent.

Equal target masks answer whether methods were scored on the same forecasting events. They do not equalize historical information. Daily can consult yesterday’s value for every target, whereas the neural models receive six hours of recent history. The comparison

concerns these two information strategies, as well as the implementation used to map recent history to a forecast.

### D. Metrics and analysis hierarchy

For a matched target set  $\mathcal{I}$  of size  $n$ , we calculate

$$\text{RMSE} = \sqrt{n^{-1} \sum_{i \in \mathcal{I}} (\hat{y}_i - y_i)^2}, \quad (1)$$

$$\text{MAE} = n^{-1} \sum_{i \in \mathcal{I}} |\hat{y}_i - y_i|, \quad (2)$$

$$\text{nRMSE} = \text{RMSE} / (y_{\max, \text{Train}} - y_{\min, \text{Train}}), \quad (3)$$

$$\text{Skill}_b = 1 - \text{RMSE}_{\text{model}} / \text{RMSE}_b. \quad (4)$$

Positive skill favors the neural model. The Train range, not DC capacity or Test range, normalizes AC errors. Bias and  $R^2$  are also reported. Negative  $R^2$  is retained; zero target variance makes  $R^2$  undefined, and zero reference RMSE makes skill undefined without an epsilon adjustment.

The external primary analysis reports the Inverted-variate model’s three-seed mean and sample SD for each site, horizon, and scope against both references. It was selected from Alice’s frozen 12/24 primary wins and best mean rank before external Test prediction. Its Alice results are descriptive focal results, not a retrospective pre-specification. Secondary analyses report all four models, individual seeds, within-setting ranks, and full/daylight differences. Deterministic baselines do not have three independent seed replicates.

The post hoc descriptive envelope takes the lowest three-seed mean neural RMSE separately for each matched comparison. It is a favorable upper bound for the evaluated neural group, not a prespecified or deployable model. SD uses  $\text{ddof} = 1$  and describes training randomness. Overlapping origins, shared targets, horizons, and scopes produce dependent summaries; win counts are not independent tests, and no unplanned significance test is attached. Cross-setting tables keep the original 24 and external site-specific eight-comparison groups distinct.

## V. ALICE SPRINGS RESULTS

### A. Primary accuracy and preserved support

Across the 24 array–horizon–scope combinations, primary RMSE wins are 12 for Inverted-variate, nine for Depthwise TCN, two for Joint-patch, one for Discrete recurrent, and zero for Last-value Persistence. The original five-method average ranks are 1.875, 2.167, 2.458, 3.667, and 4.833 in that order. Table II gives the H12 and H144 endpoints; the Supplement retains all horizons and seeds.

At full-timeline H144, the envelope RMSE is 0.776 kW for Sanyo, 0.743 kW for Hanwha, and 0.789 kW



TABLE II. Primary Test Train-range nRMSE on horizon-specific valid origins. The envelope is the post hoc minimum three-seed mean among the four compact implementations and is a descriptive upper bound, not a prespecified model; the complete model-by-seed table is in the Supplementary Material.

| Array  | Scope    | Horizon | Envelope member    | Envelope nRMSE | Last-value nRMSE | RMSE skill |
|--------|----------|---------|--------------------|----------------|------------------|------------|
| Sanyo  | Full     | H12     | Discrete recurrent | 0.0342         | 0.0766           | 0.553      |
| Sanyo  | Full     | H144    | Inverted-variate   | 0.1280         | 0.4506           | 0.716      |
| Sanyo  | Daylight | H12     | Depthwise TCN      | 0.0480         | 0.1101           | 0.564      |
| Sanyo  | Daylight | H144    | Inverted-variate   | 0.1745         | 0.6056           | 0.712      |
| Hanwha | Full     | H12     | Depthwise TCN      | 0.0301         | 0.0745           | 0.595      |
| Hanwha | Full     | H144    | Depthwise TCN      | 0.1275         | 0.4394           | 0.710      |
| Hanwha | Daylight | H12     | Depthwise TCN      | 0.0396         | 0.1078           | 0.633      |
| Hanwha | Daylight | H144    | Depthwise TCN      | 0.1782         | 0.5614           | 0.683      |
| Qcells | Full     | H12     | Inverted-variate   | 0.0542         | 0.0780           | 0.305      |
| Qcells | Full     | H144    | Depthwise TCN      | 0.1307         | 0.4005           | 0.674      |
| Qcells | Daylight | H12     | Inverted-variate   | 0.0756         | 0.1129           | 0.330      |
| Qcells | Daylight | H144    | Depthwise TCN      | 0.1909         | 0.6068           | 0.685      |

for Qcells, against Last-value errors of 2.729, 2.561, and 2.418 kW. The focal Inverted-variate model improves on Last-value in all 24 comparisons. Individual secondary implementations need not do so: at Qcells H12, Last-value RMSE is 0.471 kW and Inverted-variate reaches 0.327 kW, while two other implementations have larger errors than Last-value.

Horizon-specific support preserves 6,463 Qcells H12 origins, 77,556 full-timeline points, and 36,504 daylight points. H144 has 2,996 origins. These counts are identical across competing methods at the corresponding horizon; short-horizon eligibility does not require an uninterrupted twelve-hour target trajectory.

## B. Daily recurrence and the neural envelope

Figure 2 displays the complete 24-comparison envelope evidence. Daily Persistence has lower matched RMSE in 22 comparisons. The two neural-envelope wins are Hanwha H12: Depthwise TCN reaches 0.176 kW against Daily’s 0.185 kW on the full timeline and 0.231 against 0.273 kW in daylight. The focal Inverted-variate model has no Daily wins in this Alice comparison. This distinction between a named model and a Test-selected envelope is retained throughout the cross-setting analysis.

Rankings also vary among the co-located arrays. Joint-patch leads both Sanyo H48 scopes; Discrete recurrent leads Sanyo H12 full; Depthwise TCN leads most Hanwha conditions; and Inverted-variate leads most Qcells conditions and Sanyo H96/H144. Full/daylight differences matter even with the same history and weights. The original accuracy–efficiency comparison is retained in the Supplement: Joint-patch has the lowest measured batch-one latency on the documented laptop GPU, while Inverted-variate has the best average accuracy rank. These timings describe the 17-channel implementations in that environment.

## VI. EXTERNAL-SITE REPLICATION RESULTS

### A. Prespecified model against both references

Table III reports all sixteen external primary combinations. Figures 3 and 4 place the focal model’s Alice results beside its prespecified external results, with the input-regime boundary visible. Inverted-variate improves on Last-value in all eight Yulara and all eight NIST comparisons. At Yulara, Last-value skill increases from 17.34% at H12/full to 67.10% at H144/full; the corresponding NIST values are 17.93% and 43.24%. The positive signs recur across both scopes and all horizons.

The external Daily comparison differs from Alice. Yulara has eight positive primary skills, and NIST has seven. At Yulara H12, Daily skill is 36.28% full and 37.28% daylight. At H96 it is 2.71% and 6.10%, and at H144 it is 1.62% and 4.50%. Thus a positive sign at the longer horizons corresponds to a smaller error reduction than at H12. At NIST H144/full, skill is  $-6.47\% \pm 1.65\%$ : the primary model’s mean RMSE is 6.47% higher than Daily’s on their matched targets. H144/daylight instead gives  $13.20\% \pm 1.49\%$  skill. The scope changes the comparison even at the same site and horizon.

The Supplement provides absolute RMSE and MAE, all seeds, bias,  $R^2$ , and sample counts. Train-range nRMSE in Table III makes the within-site error scale explicit without treating different facilities’ kW errors as interchangeable. Seed SDs quantify the observed training variability around each result rather than confidence intervals over future seasons.

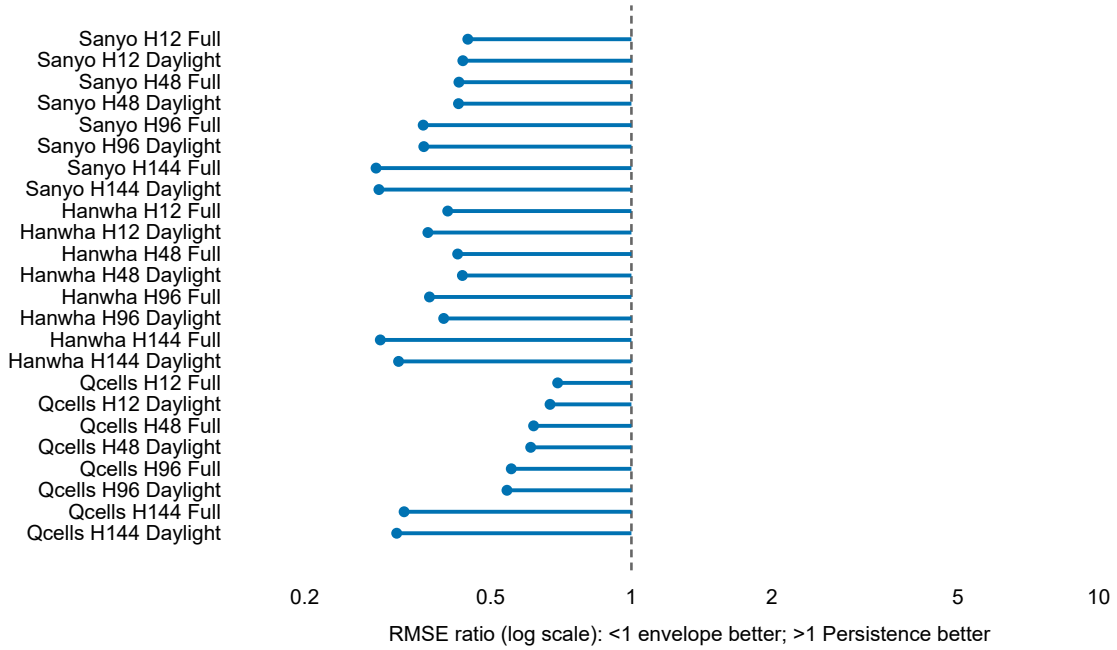
### B. Architecture rankings and descriptive envelope

Yulara retains Inverted-variate as the lowest-RMSE neural implementation in all eight primary combinations. At NIST, Discrete recurrent leads all four full-timeline comparisons, Inverted-variate leads H12/daylight, and

## The baseline choice reverses the practical conclusion

(a) Envelope / Last-value Persistence

Neural envelope lower than Last-value: 24/24.



(b) Envelope / Daily Persistence (matched points)

Daily Persistence lower than neural envelope: 22/24.

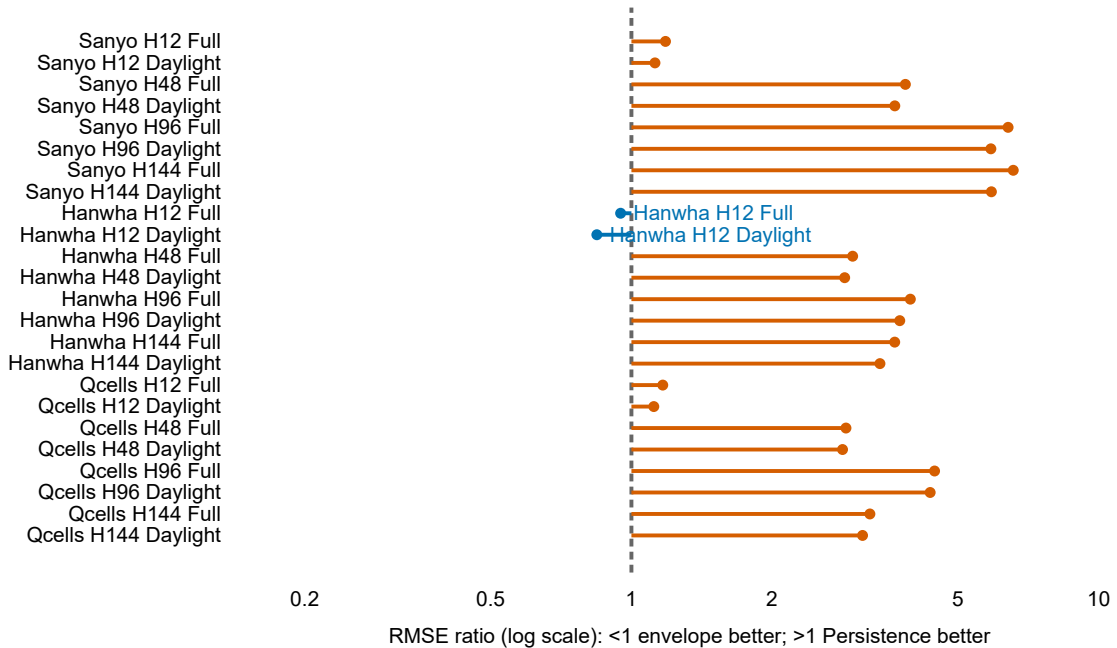


FIG. 2. Complete Alice Springs envelope-to-reference RMSE ratios. The post hoc best-of-four neural envelope is compared with Last-value on primary support and Daily on its separately matched support. Ratios below one favor the envelope; those above one favor the reference. The two Daily exceptions are Hanwha H12 full and daylight.

TABLE III. External prespecified Inverted-variate results: three-seed mean  $\pm$  sample SD. nRMSE and skill are percentages; Daily skill uses its own common target intersection.

| Site        | Horizon | Scope    | Train-range nRMSE | Last-value skill | Daily-matched skill |
|-------------|---------|----------|-------------------|------------------|---------------------|
| Yulara      | H12     | full     | $9.36 \pm 0.03$   | $17.34 \pm 0.26$ | $36.28 \pm 0.20$    |
| Yulara      | H12     | daylight | $12.72 \pm 0.06$  | $18.73 \pm 0.36$ | $37.28 \pm 0.28$    |
| Yulara      | H48     | full     | $12.16 \pm 0.13$  | $49.24 \pm 0.53$ | $17.25 \pm 0.86$    |
| Yulara      | H48     | daylight | $16.06 \pm 0.22$  | $48.58 \pm 0.70$ | $20.83 \pm 1.08$    |
| Yulara      | H96     | full     | $14.28 \pm 0.05$  | $61.32 \pm 0.14$ | $2.71 \pm 0.34$     |
| Yulara      | H96     | daylight | $19.03 \pm 0.14$  | $56.10 \pm 0.31$ | $6.10 \pm 0.67$     |
| Yulara      | H144    | full     | $14.37 \pm 0.14$  | $67.10 \pm 0.32$ | $1.62 \pm 0.97$     |
| Yulara      | H144    | daylight | $19.28 \pm 0.20$  | $59.46 \pm 0.42$ | $4.50 \pm 0.99$     |
| NIST Ground | H12     | full     | $6.91 \pm 0.04$   | $17.93 \pm 0.52$ | $57.73 \pm 0.28$    |
| NIST Ground | H12     | daylight | $11.01 \pm 0.04$  | $20.24 \pm 0.26$ | $59.01 \pm 0.13$    |
| NIST Ground | H48     | full     | $11.05 \pm 0.10$  | $41.16 \pm 0.51$ | $32.09 \pm 0.58$    |
| NIST Ground | H48     | daylight | $17.22 \pm 0.39$  | $37.98 \pm 1.39$ | $36.07 \pm 1.41$    |
| NIST Ground | H96     | full     | $15.59 \pm 0.29$  | $42.18 \pm 1.06$ | $3.88 \pm 1.80$     |
| NIST Ground | H96     | daylight | $21.58 \pm 0.30$  | $39.25 \pm 0.85$ | $20.22 \pm 1.12$    |
| NIST Ground | H144    | full     | $17.29 \pm 0.27$  | $43.24 \pm 0.87$ | $-6.47 \pm 1.65$    |
| NIST Ground | H144    | daylight | $23.49 \pm 0.40$  | $37.80 \pm 1.06$ | $13.20 \pm 1.49$    |

TABLE IV. Mean neural-only primary RMSE ranks. Alice pools ranks across three co-located arrays, not independent sites; external sites remain separate.

| Setting       | Combinations  | Recurrent | Inverted | Joint-patch | TCN   |
|---------------|---------------|-----------|----------|-------------|-------|
| Alice Springs | 24 (3 arrays) | 3.583     | 1.875    | 2.375       | 2.167 |
| Yulara        | 8             | 3.500     | 1.000    | 2.500       | 3.000 |
| NIST Ground   | 8             | 1.750     | 2.250    | 4.000       | 2.000 |

TABLE V. Model/envelope comparisons won on matched targets. Wins are dependent descriptive summaries, not independent significance tests. The envelope is a favorable upper bound for the neural group.

| Setting       | Selection status     | Versus Last-value | Versus Daily |
|---------------|----------------------|-------------------|--------------|
| Alice Springs | Descriptive focal    | 24/24             | 0/24         |
| Alice Springs | Post hoc envelope    | 24/24             | 2/24         |
| Yulara        | Prespecified primary | 8/8               | 8/8          |
| Yulara        | Post hoc envelope    | 8/8               | 8/8          |
| NIST Ground   | Prespecified primary | 8/8               | 7/8          |
| NIST Ground   | Post hoc envelope    | 8/8               | 7/8          |



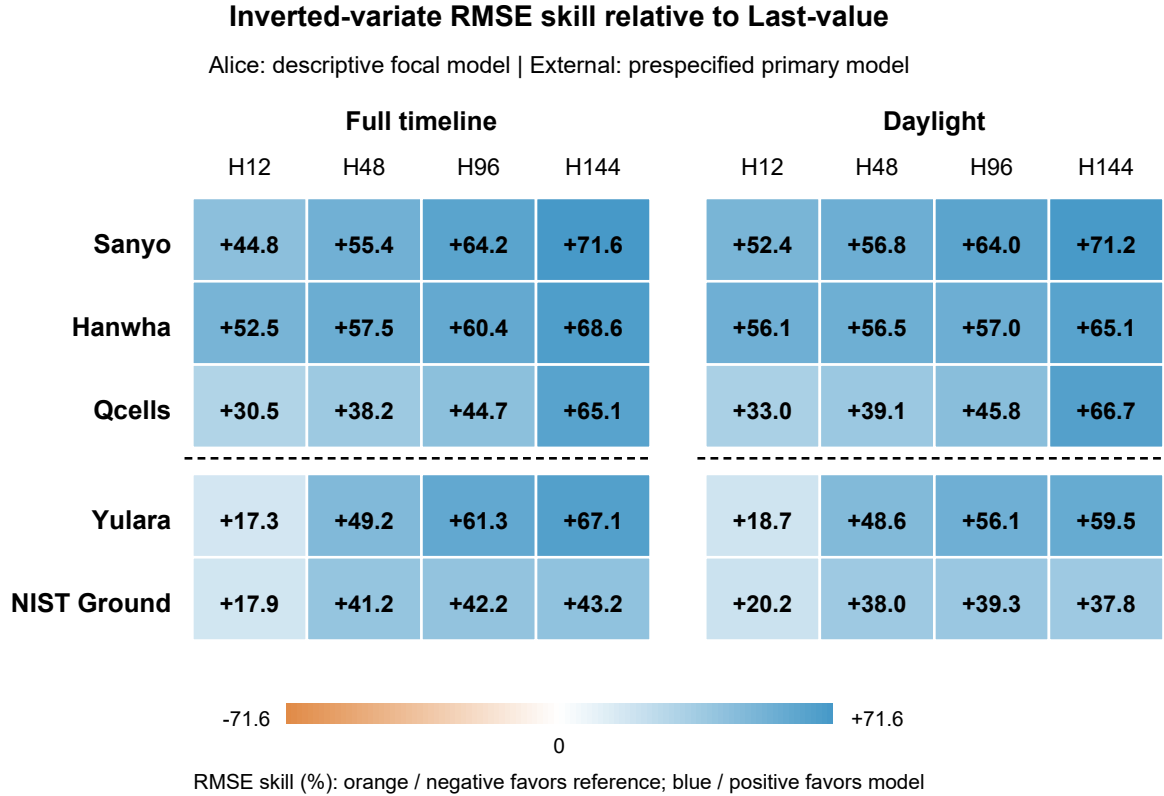


FIG. 3. Inverted-variate RMSE skill (%) against Last-value Persistence on primary matched targets. Alice rows are descriptive focal-model comparisons; external rows are the prespecified primary analysis. Signed cell values retain readability in grayscale. The dashed boundary separates the 17-channel co-located comparison from seven-channel external replications.

Depthwise TCN leads H48, H96, and H144/daylight. NIST mean neural ranks are 1.75 for recurrent, 2.00 for TCN, 2.25 for Inverted-variate, and 4.00 for Joint-patch (Table IV). Figure 5 preserves the three separate Alice array rows and each external facility.

Programmatic envelope comparisons give eight Yulara wins and seven NIST wins against Daily (Table V); NIST H144/full remains the Daily win after favorable neural-member selection. The envelope and primary model therefore have the same external win counts, although the envelope can use different implementations and errors. These counts do not make the two analyses equivalent. The prespecified model provides the primary replication result, while the envelope describes the best observed mean within the evaluated neural group.

## VII. CROSS-SETTING INTERPRETATION

### A. Improvement beyond continuity and daily recurrence

The repeated Last-value advantage supports a specific conclusion: multivariate recent history improves on holding origin power constant in these settings. Last-value has no mechanism for adapting its trajectory through a

day–night transition. A learned trajectory can summarize recent power and irradiance behavior and represent changing production levels. This explains why the reference is informative at short horizons yet increasingly demanding to interpret as an operational alternative over twelve hours. It does not identify which input or network component produces the improvement; that would require a separately designed ablation.

Daily uses a different information strategy. Yesterday’s power supplies a timestamp-specific diurnal trajectory across the entire horizon. Its broad Alice advantage shows the usefulness of daily recurrence during that August Test period. The external results show that the same pattern does not hold uniformly for Yulara and NIST in October–December. Differences in day-to-day weather, operational state, seasonal timing, and the match between yesterday’s and today’s production are plausible explanations. They are hypotheses consistent with the observations, not demonstrated mechanisms: the benchmark does not isolate their contributions.

The two external sites also differ from each other. Yulara’s primary Daily skill approaches zero at longer horizons while retaining a positive mean. NIST’s H144 result changes sign between full and daylight. Comparing both scopes makes this distinction visible instead of allowing

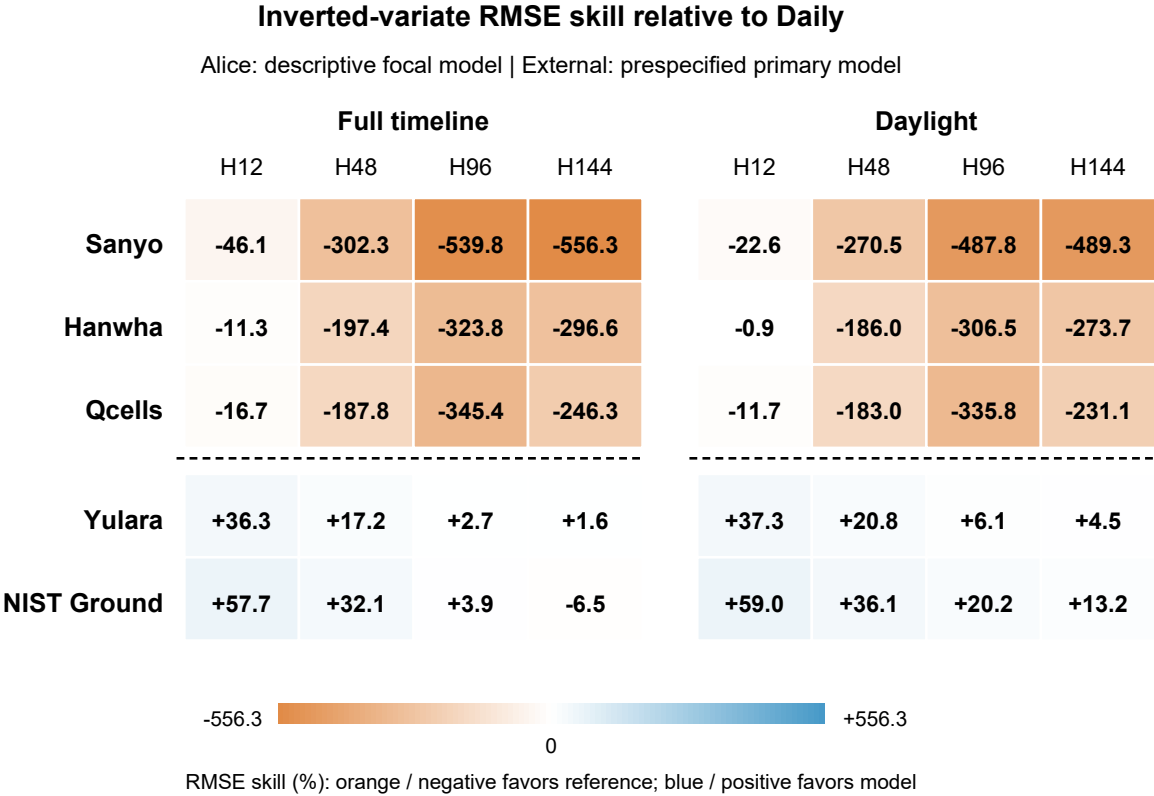


FIG. 4. Inverted-variate RMSE skill (%) against Daily Persistence on the common Daily-valid point intersection. Selection status and input-regime grouping match Fig. 3. Positive values favor Inverted-variate; negative values favor Daily. All Alice cells and NIST H144/full favor Daily, while the other external cells favor the prespecified model.

the many low-production targets to define the complete narrative. Because each comparison uses shared points, the sign change describes errors on a different evaluation scope rather than a method-specific deletion rule.

**B. Rank changes and information regimes**

At NIST, full-timeline and longer daylight targets favor different compact implementations. Full-timeline loss includes low and negative meter values as well as daytime production; daylight concentrates on targets above the Train-derived threshold. Different error distributions across these regions can yield different aggregate ranks. Recurrent and depthwise structures may represent these trajectories differently, but the observed ranking is not evidence that one structural property caused the outcome. Parameter count, optimization dynamics, and the sampled operating distribution also differ in their effects.

The seven-channel protocol enables a common-input external comparison by using fields available at both facilities. It does not reproduce Alice’s richer sensor set or transfer its fitted weights. Years, capacities, local coordinates, target-validity rules, and evaluation months also differ. Consequently, Table IV averages

ranks within each setting, and Table V keeps the settings separate. A pooled kW ranking or an undifferentiated forty-comparison win total would conceal these distinctions.

**C. What matched support establishes**

Exact target matching removes one important ambiguity: a baseline cannot gain or lose apparent accuracy simply because unavailable lag points were removed from its own score alone. Neural, Last-value, and Daily errors on the Daily intersection describe the same target events. Horizon-specific eligibility likewise keeps short-horizon forecasts whose later targets are missing without granting any method additional scored points.

Matching does not equalize the age or amount of historical information. A six-hour learned history and a target-aligned 24-hour lag are distinct strategies, and their relative value depends on the setting. Nor does matching turn overlapping forecast origins into independent observations. The practical evidence is a collection of well-defined conditional comparisons, with complete support counts and seed results available for inspection.

The three geographic sites broaden the evidence be-

### Mean within-array neural rank across four horizons and two scopes

1 = lowest RMSE; each rank compares four compact implementations

|                    | Discrete recurrent | Inverted-variate | Joint-patch | Depthwise TCN |
|--------------------|--------------------|------------------|-------------|---------------|
| <b>Sanyo</b>       | 3.38               | 1.88             | 2.12        | 2.62          |
| <b>Hanwha</b>      | 3.62               | 2.50             | 2.50        | 1.38          |
| <b>Qcells</b>      | 3.75               | 1.25             | 2.50        | 2.50          |
| <hr/>              |                    |                  |             |               |
| <b>Yulara</b>      | 3.50               | 1.00             | 2.50        | 3.00          |
| <b>NIST Ground</b> | 1.75               | 2.25             | 4.00        | 2.00          |

Top three rows: one co-located Alice facility (17 channels)

Bottom two rows: separate external facilities (7 channels); no pooled kW ranking

FIG. 5. Mean neural-only primary RMSE ranks across four horizons and two scopes within each target system. Rank one is best among four implementations. The three Alice rows are co-located arrays, not independent geographic replications. Gray shading and printed values encode the same ranks. No absolute kW errors are averaged across sites.

yond a single co-located facility. They reproduce the focal model’s improvement over Last-value, show a different Daily pattern outside Alice, and reveal an external ranking change. This supports reporting site, horizon, scope, and input regime alongside every headline result. Studies using future weather, probabilistic forecasts, or geographic transfer address complementary questions.<sup>15–18</sup> Their reported errors should be interpreted under their own information sets and sample definitions.

## VIII. LIMITATIONS

The evidence comprises three geographic sites. Alice’s three arrays share a facility and weather context, and the external held-out performance splits cover October–December rather than all seasons. Alice uses 2018 data and 17 channels; the external experiments use 2017 and seven channels. These differences and the separately fitted weights define external-site replication, not zero-shot transfer, a uniform-input cross-climate experiment, or an isolated technology effect.

The complete NIST 2017 operation log is unavailable, and detailed Yulara field metadata remain partial. Finite external negative values and possible unusual operating states are retained under the frozen rules. Native Yulara

GHI is scaled using Train data without cross-site absolute irradiance interpretation. Conservative availability rules address timestamp-interval ambiguity but do not supply missing sensor metadata. Test records were inspected for data quality during development and are described as held-out performance splits.

The four networks are compact project implementations; their rankings do not establish the ordering of full official architectures. Three-seed SD measures training randomness, not temporal sampling uncertainty. Correlated windows and comparison counts do not support unplanned significance claims. Daylight is a true-target descriptive scope, and Daily’s 24-hour information differs from the neural six-hour history. The study evaluates deterministic history-only forecasts; future weather and probabilistic decision value are separate objectives.

## IX. CONCLUSION

The two-level benchmark distinguishes improvement beyond local continuity from improvement beyond daily recurrence. At Alice Springs, the focal Inverted-variate model improves on Last-value across all 24 comparisons, while Daily has lower error than the favorable post hoc neural envelope in 22. In the independently fitted external experiments, the prespecified model improves on

Last-value in all sixteen comparisons and on matched Daily in fifteen. Yulara preserves its leading neural rank, whereas NIST favors recurrent or depthwise implementations in several scopes.

These findings make the conditioning variables part of the result: site, horizon, scope, and historical information determine how a model's advantage should be interpreted. Train-only transformations, explicit availability times, split-local windows, and elementwise target matching provide a common basis for that interpretation across both input regimes.

## SUPPLEMENTARY MATERIAL

The Supplement contains complete Alice evidence, external field and time audits, all 96 support groups, the 24-run matrix and checkpoint records, per-seed RMSE, complete mean/sample-SD errors, metadata limits, model configurations, and the descriptive envelope. Full per-seed metric CSVs accompany the code rather than being expanded into thousands of PDF rows.

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## CONFLICT OF INTEREST

**Author confirmation required.** Draft: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## ETHICS APPROVAL

**Author confirmation required.** Draft: This study analyzes publicly accessible photovoltaic and meteorological records and involves no human participants or animals; ethics approval was not applicable.

## AUTHOR CONTRIBUTIONS

**Author confirmation required.** Draft: Jiangkun Zhu: Conceptualization, Methodology, Software, Validation,

Formal analysis, Investigation, Data curation, Writing – Original Draft, Visualization. Mengling Yang: Validation, Writing – Review & Editing. Zhicong Chen: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – Review & Editing. Lijun Wu: Supervision, Resources, Project administration, Funding acquisition, Writing – Review & Editing.

## DECLARATION OF AI-ASSISTED TOOLS

**Author confirmation required.** The exact tool/version record and any journal-required disclosure wording remain pending author confirmation; no model designation is inferred.

## DATA AVAILABILITY

**Author confirmation required.** Draft: Source observations are available from the official DKA Solar Centre services for Alice Springs and Yulara and from the NIST PV data service.<sup>10,13,34</sup> Reproduction requires downloading data from those providers and complying with their terms. The code repository does not redistribute these raw observations. The final public code location and release details remain subject to author confirmation.

## CODE AVAILABILITY

**Author confirmation required.** Draft: A dedicated public release of evaluation code, aggregate metrics, figure generation, and ordinary protocol tests is planned after manuscript approval and license selection. Raw data, checkpoints, and prediction arrays will not be redistributed. The final repository URL must be confirmed before submission.

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